CLEANBOT AI

Single-Robot Office Cleaning Management System

MASTER PROMPT FOR GOOGLE AI STUDIO BUILD MODE

Initial scope: one office cleaning robot • Simulation-first development • Future hardware integration

1. PROJECT OVERVIEW

Build a complete, professional office cleaning robot management system called **CleanBot AI**. It manages **one autonomous cleaning robot** that maintains an office environment.

Cleaning areas: office floors and corridors, desks and workstations, tables and meeting rooms, reception areas, conference rooms, and shared spaces. The system must:

1. Clean floors, corridors, desks, and tables.
2. Detect and classify wet and dry dirt (simulated initially).
3. Navigate office rooms and avoid obstacles.
4. Monitor battery and waste container levels.
5. Return automatically to its service station when the container is full or the battery is low.
6. Notify the operator when it needs assistance.
7. Resume unfinished tasks after servicing.
8. Record cleaning activity and generate reports.

Scope rules: exactly one registered robot, one robot dashboard, one navigation map, one active task at a time (others wait in a queue), and one assigned service station. There is no fleet management and no multi-robot coordination. Keep the data model clean so multi-robot support could be added later, but do not build it now.

2. TECHNOLOGY STACK

- **Frontend:** React, TypeScript, Tailwind CSS, Lucide React icons, Recharts, Framer Motion.
- **Backend:** Node.js and Express (or equivalent), REST APIs, WebSockets for real-time updates.
- **Database:** Firebase Firestore or another persistent database. Store users, the robot, office layout, tasks, sensor readings, alerts, and history.
- **Robot communication:** MQTT-ready architecture with WebSocket support. Keep the simulation layer and the physical hardware layer fully separate behind a common interface.

3. AUTHENTICATION AND ROLES

Implement secure authentication and role-based access control throughout.

- **Administrator:** manage users and permissions; configure the robot, office layout, zones, and station; create schedules; view reports and analytics; change settings.
- **Operator:** monitor the robot; create cleaning requests; start, pause, resume, and stop permitted tasks; empty the waste container; respond to alerts; review progress.

4. OFFICE MANAGEMENT

Administrators can register the building, create floors, add and edit rooms, register desks and tables, define corridors and shared spaces, create cleaning zones, configure restricted areas, and set cleaning priorities. The single robot covers all zones. Implement full CRUD.

5. ROBOT MANAGEMENT (SINGLE ROBOT)

One robot record with: Robot ID, name, model, type, current location, battery %, container capacity, status, current task, assigned service station, and last communication time.

Statuses: Available, Cleaning, Returning to Station, Charging, Waiting for Operator, Paused, Error, Offline.

Administrators can edit and configure the robot. Provide a registration or setup screen shown only if no robot exists yet, and prevent adding a second robot.

6. AUTONOMOUS CLEANING OPERATIONS

Request types: floor only, table only, desk only, combined, or complete room.

Each task includes: Task ID, cleaning location, cleaning type, priority, start time, expected duration, status, and progress %. The robot runs **one task at a time**. Additional tasks go into a priority-ordered queue.

Implement create, queue, schedule, start, pause, resume, cancel, and complete.

7. DIRT DETECTION (SIMULATED)

Categories: dust, sand, paper, food particles, liquid spills, other debris. Classifications: **DRY, WET, MIXED, UNKNOWN**.

Show dirt type, location, time, confidence (when available), and recommended cleaning method. Cleaning methods depend on the robot's actual mechanisms and surface compatibility.

Use simulated detection initially and prepare camera and dirt-sensor integration points. **Label simulated detection clearly and never present it as real sensor or AI detection.**

8. NAVIGATION AND OFFICE MAP

Interactive map showing floors, rooms, corridors, desks, tables, restricted areas, obstacles, the robot's position, the planned route, completed areas, and pending areas.

In simulation, animate the robot moving and cleaning, simulate obstacle detection and rerouting, and block entry to restricted zones. Prepare for future LiDAR, camera, and localization integration.

9. WASTE CONTAINER MONITORING AND FULL-CONTAINER WORKFLOW

Show container capacity, percentage, status, and full-container alerts. When the configured maximum is reached:

1. Stop cleaning safely.
2. Save completed tasks.
3. Save current task progress.
4. Record unfinished tasks.
5. Return to the assigned service station.
6. Notify the operator.

7. Set status to Waiting for Operator.
8. Require the operator to empty the container.
9. Verify emptiness (sensor, or explicit operator confirmation in simulation).
10. Resume the first unfinished task.
11. Do not repeat completed work.

The robot must **not** resume until the container is confirmed empty and all safety conditions are met.

10. BATTERY AND CHARGING

Battery monitoring, low-battery alerts, configurable return thresholds, automatic return-to-station, charging progress, and automatic availability after charging. Save unfinished work before returning, and resume only when battery level and safety conditions allow.

11. SERVICE STATION

One station module showing station ID, location, robot connection status, charging status, arrival and departure history, container-emptying status, and maintenance needs.

12. NOTIFICATIONS AND ALERTS

Working notification center with alerts for task completion, new requests, task start, low battery, full container, return to station, charging complete, navigation problems, obstacles, communication failure, operator intervention, and maintenance. Include history, timestamps, priority, and read/unread status.

13. CLEANING HISTORY AND REPORTS

Record each task's robot ID, location, type, start and end times, duration, detected dirt categories, result, and operator interventions. Filter by date, room, zone, cleaning type, and status. Provide exportable reports (CSV and PDF).

14. ANALYTICS

Interactive charts for total, completed, pending, and failed tasks; cleaning time; robot uptime and utilization; performance by room and zone; container usage; battery performance; and daily and weekly activity. **Calculate all statistics from stored records, with no hardcoded values.**

15. MAIN DASHBOARD

- **Summary cards:** robot status, battery %, container %, completed tasks, pending tasks, active alerts.
- **Panels:** live robot status, interactive map, current task and progress, battery gauge, container gauge, recent alerts, task queue.
- **Charts:** daily activity, weekly performance, task distribution.

16. NAVIGATION (SIDEBAR)

Dashboard; Robot Management; Office Management; Floor and Room Management; Cleaning Zones; Cleaning Requests; Active Cleaning Task; Dirt Detection; Navigation Map; Container Monitoring; Battery and Charging; Service Station; Notifications; Cleaning History; Analytics and Reports; User Management; Settings. **Every item must open a working page.**

17. SETTINGS

Cleaning priorities, container thresholds, battery thresholds, operating schedule, notification preferences, office configuration, robot communication, simulation mode, and hardware integration.

18. SIMULATION MODE

A full simulation with a sample office (multiple rooms, desks, tables, corridors, restricted areas, obstacles), **one simulated robot**, simulated battery and container levels, simulated dirt detection, navigation, tasks, and one service station.

Controls: **Start, Pause, Reset**, plus a speed control. Provide buttons to force a full container or low battery so the return-and-resume workflows can be tested. Label all simulated data clearly.

19. PHYSICAL HARDWARE INTEGRATION

Build a modular hardware abstraction layer for cameras, LiDAR, ultrasonic sensors, dirt sensors, battery sensors, container sensors, motors and actuators, navigation controller, and an MQTT gateway. Use one **RobotDriver** interface with two implementations: **SimulatedRobotDriver** and **PhysicalRobotDriver** (stub).

Do not claim the app controls a physical robot until hardware, communications, and safety systems are connected and tested.

20. UI/UX

Modern industrial IoT enterprise look: dark navy and white surfaces, blue and cyan accents, green for success, orange for warnings, red for critical alerts. Clean typography, interactive cards, modern tables, clear charts, smooth transitions, consistent icons, and accessible controls. Fully responsive on desktop, tablet, and mobile.

21. FUNCTIONAL REQUIREMENTS

This is a working application, not static pages. Every button works, forms validate, CRUD persists to the database, and dashboard stats update from real data. Search, filters, sorting, status updates, task progress, event-driven notifications, error handling, loading states, and empty states must all work. Enforce authentication and permissions.

22. DEVELOPMENT AND TESTING

Use reusable components, clear folder structure, separate API layers, and consistent data models.

Test: every page and nav item; every button and form; all CRUD; persistence; permissions; robot simulation; task execution and queueing; the full-container return-and-resume workflow; the low-battery return-and-charge workflow; notifications; reports and analytics. Fix all runtime errors.

Produce a **functional verification report** with: feature name, implementation status, test result, remaining issues. Do not mark anything complete unless it is implemented and tested. Clearly separate **fully implemented software**, **simulated functionality**, and **features awaiting physical hardware**.

23. FINAL OBJECTIVE

Deliver a functional, maintainable, expandable single-robot office cleaning platform. The first version must run in simulation and demonstrate the complete workflow: office setup, robot setup, autonomous cleaning,

container monitoring, service-station return, operator servicing, and unfinished-task resumption. Prioritize working functionality, reliable data, safe robot workflows, and a professional experience over decorative elements.